

## Risk policy memo — aluminium scrap import desk (SIM)

**To:** Risk Committee (SIM) · **From:** Head of Desk (SIM), Meridian Non-Ferrous Trading (SIM) · **Date:** 7 November 2022 · **Re:** limits for the next book, set against what the Mar-Oct 2022 book actually did

**ACADEMIC SIMULATION — not actual trades.** Book P&L was ₹192.1 m at 31-Oct-2022, but it is **not sign-robust**: –₹105.4 m to +₹334.3 m across the registered domestic anchor-premium grid (break-even –38,702 ₹/t inside it). So every limit below is sized on the risk the book ran, not on what it earned. Evidence is from the published tables; checks that include the MCX leg skip the 16 MCX exit/roll position dates P3 flags. Hedge effectiveness is measured on a unit-beta MCX proxy, so the MCX basis is under-represented in every hedge number here. The third-party mirror's beta is 0.75 on weekly closes and 0.45 on daily data, which the MCX evening close biases down: mean GARCH VaR is ₹8.5 m at the weekly beta and ₹17.3 m at the daily one (the pessimistic end, not a hedge ratio anyone would use), against ₹4.27 m on the proxy.

| Area                | Limit / rule proposed                                                                                                                                                                                                                                                                            | Book evidence (2022)                                                                                                                                                                                                                                                                                                      | Book vs limit → remediation                                                                                                                                                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Position</b>     | Gross physical LME exposure ≤ 5,000 MT and ≤ USD 15 m                                                                                                                                                                                                                                            | Peak 4,450 MT / USD 14.5 m on 21-Apr; unsold cargo 4,980 MT on 6-Apr                                                                                                                                                                                                                                                      | Within by construction — the limit was set just above the book's own peak, so this is not evidence; not raised, because that size already used the whole working-capital line (Liquidity row)                                                                                        |
| <b>Unhedged</b>     | Net LME delta ≤ 1,000 MT and ≤ USD 3 m; net ≤ 25 % of physical once physical ≥ 1,000 MT                                                                                                                                                                                                          | Absolute net: 95th pct 967 MT; max 1,817 MT (USD 6.36 m) on 8-Mar, the naked session before T01's hedge                                                                                                                                                                                                                   | MT breached 5 days (8-Mar-6-Jul); USD breached 2 days (8-Mar-14-Apr); % breached 26 of 99 days (ratios drift between re-sizings) → rebalance within 2 sessions; no purchase before its hedge can be placed                                                                           |
| <b>Hedge</b>        | MCX ratio 0.80–1.10 of LME-equivalent physical at every decision; roll ≥ 7 trading days before expiry, roll gains never budgeted (proxy curve is always contango); FX forwards ≥ 90 % of each fixed USD payable, on physical + forwards only                                                     | 8 tickets filled 0.899–1.002; T04 at 0.50; 13 of 13 rolls by the deadline; FX 100 % except T06 60 %                                                                                                                                                                                                                       | T04 and T06 breach → re-size, or a committee exception signed before the trade                                                                                                                                                                                                       |
| <b>Stop-loss</b>    | Ticket –₹10 m from its own peak: same-day review. Ticket –₹20 m: hedge to 1.0, no same-grade additions. Book –₹40 m from peak: no new purchase until the committee reviews                                                                                                                       | Review level hit by T08 –₹26.1 m, T05 –₹14.0 m, T07 –₹11.9 m, T04 –₹10.9 m; book –₹49.9 m (9-Jun→8-Aug), past –₹40 m from 1-Aug                                                                                                                                                                                           | T08 hit its hard stop on 1-Aug, the book its stop from 1-Aug; T09 bought 3-Aug would have been refused (same grade as T08). Honest cost: T09 finished +₹10.0 m on base marks. The levels were chosen with these drawdowns in view, so which tickets trip is illustration, not a test |
| <b>VaR</b>          | 1-day 95 % VaR ≤ ₹10 m on the larger of GARCH(1,1) and 250-day historical (docs/40: the historical is the floor, because GARCH decayed below it into the April-May crash). Covers LME, FX and freight only: grade spread (daily σ ₹2.2 m) and the cash–3M spread (memo VaR +18 %) sit outside it | 120 position days: GARCH mean ₹4.27 m, max ₹43.5 m on 9-Mar                                                                                                                                                                                                                                                               | Breached once (9-Mar) on clean days (the naked session); 5 days counting roll-day artefacts → cut the driving position the same day                                                                                                                                                  |
| <b>Counterparty</b> | Band from the illustrative score: A max use 100 %, B 90 %, C 75 % + security + advances ≤ 0.5x limit, D line cut to 50 %, advance/CAD only; never lift a hedge before an advance lands                                                                                                           | BUY_RJK_01 band D (model PD 41.8 %, synthetic-data model); advances > 1.0x its ₹120 m line on 37 days, peak 1.83x, contracted 2.57x; 5 of 10 bookings outside the bands                                                                                                                                                   | Breached → BUY_RJK_01 line ₹120 m → ₹60 m, advance or CAD only; BUY_JNPT_01 band C: freeze; BUY_MUN_01 band C: freeze                                                                                                                                                                |
| <b>Freight</b>      | FOB freight fixed ≤ 10 trading days after purchase; unhedged with a stop ≤ fixture + 20 %; no proxy hedge (no instrument tracks the India lanes; the two lanes are one factor)                                                                                                                   | Fixed after T01 8, T04 6, T07 2 days; stops +18.4 %, +12.7 %, +14.2 %; freight VaR mean ₹0.05 m; hypothetical +40 % spike costs T01 –₹8.55 m                                                                                                                                                                              | Within                                                                                                                                                                                                                                                                               |
| <b>Liquidity</b>    | Working-capital line ₹1,000 m and LC line ₹1,000 m (sized on the Feb-2022 plan). Funding need + max(₹150 m, 99 % 10-day margin-at-risk at 15 % IM) ≤ line, checked daily; no LC opened that takes the LC line over limit                                                                         | Peak need ₹1,179.0 m on 20-Jul; LC peak ₹1,414.8 m on 13-Jun; margin-at-risk peak ₹304.4 m on 21-Apr; crash fortnight (22-Apr–9-May) actually paid in +₹65.0 m of VM, but LME +23 % against the MCX shorts calls ₹181.8 m more and takes headroom from ₹0.2 m to –₹181.6 m while the cargo gains ₹427 m of mark, not cash | Buffer rule breached 30 days (10-May–5-Aug); line breached 19 days (16-May–5-Aug); LC breached 17 days (28-Apr–21-Jun) → pre-agree ad-hoc limits of ₹500 m (WC) and ₹500 m (LC) for peak months, or stagger usance maturities and purchases                                          |

**Decisions requested.** (1) Adopt the limits above for the next book. (2) Approve the BUY\_RJK\_01 line cut and the advance-before-hedge-lift rule now. (3) Mandate the treasury to seek the ad-hoc lines before the next book's peak months; until they are in place, no ticket is signed that the daily liquidity rule shows breaching the line within its cash cycle.

**What these limits cannot see.** The position, unhedged and stop levels were set with the 2022 book in view, so where the book lands against them is illustration, not evidence; the VaR limit excludes grade spread, the book's largest bucket after deal margin. The MCX series and USD/INR are proxies; grade factors and freight levels are hindsight reconstructions; credit PDs come from a model fitted to synthetic data; no drawing-power (stock and receivables) test is modelled; the stop-loss rules would not have fixed the drawdowns driven by grade spread and carry, only stopped the desk adding to them.